Parameter Monitoring with Adaptive Observers

Estimating T cells and patient-specific parameters in SARS-CoV-2

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Today: a 4-step recipe

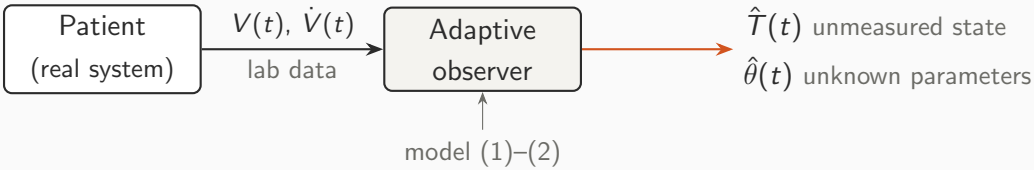
1. **Model.** Write down the dynamics and decide what is *measured*, what is *unknown*.
2. **Can we?** Observability (states) and identifiability (parameters).
3. **Rewrite.** Put the model in the state-affine form

$$\dot{x} = A(y)\,x + \beta(y) + \varphi(y)\,\theta, \qquad y = Cx.$$

4. **Design and code.** Copy the adaptive observer, check its assumptions, tune it, run it in MATLAB/Python.

Two worked examples: **Ex. 1** estimate c_T · **Ex. 2** estimate c_T and r. Both also recover the unmeasured T cells.

Why an adaptive observer?



- We can measure the **viral load** V, but not the CD8$^+$ **T cells** T that fight it.
- Parameters (killing rate, activation rate, ...) are **different for every patient**.
- An adaptive observer estimates **states and parameters at the same time**, while data arrive: the basis of a *digital twin*.

Step 1 · The model (Hernandez-Vargas & Velasco-Hernandez, 2020)

$$\frac{dV}{dt} = pV\left(1 - \frac{V}{K}\right) - c_T\, V\, T - cV \qquad (1)$$

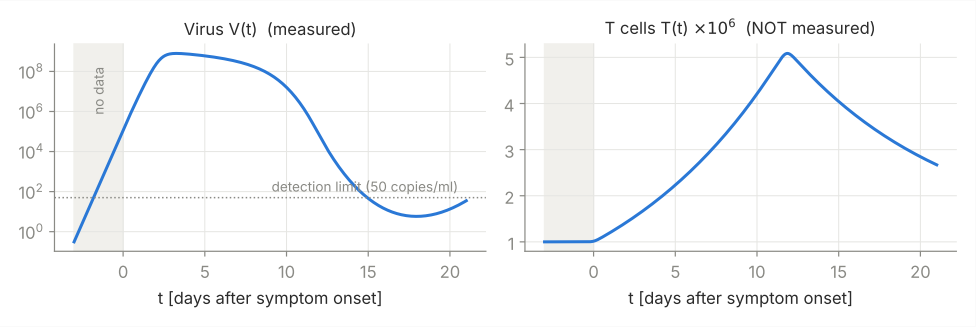
$$\frac{dT}{dt} = S_T + r\, T\, \frac{V^m}{V^m + k_T^m} - \delta_T T \qquad (2)$$

$$G(V) = \frac{V^m}{V^m + k_T^m} \quad \text{(T-cell activation)}$$

$$\alpha = p - c, \qquad \sigma = p/K, \qquad \gamma = c_T S_T$$

	meaning	"mean" patient
p	virus replication	8.57 /day
K	carrying capacity	2.35×10^9
c	virus clearance	2.4 /day
c_T	killing by T cells	1.89×10^{-6}
r	T-cell activation	0.194 /day
δ_T	T-cell death	0.1 /day
k_T, m	half-sat., Hill	1.26×10^5, 2
S_T	replenishment $= \delta_T T_0$	10^5

Our virtual patient



- **Measured:** $V(t)$ (and its derivative $\dot{V}$) from $t = 0$ (symptom onset).
- **Unknown:** $T(t)$ and the parameters c_T (Ex. 1) or c_T, r (Ex. 2).

Step 2 · Can we do it?

Observability (Theorem 1 in the paper)

With output $y = V$, stack V and $\dot{V}$ and take the Jacobian w.r.t. (V, T):

$$\frac{\partial}{\partial(V,T)}\begin{bmatrix} V \\ \dot{V} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ * & -c_T V \end{bmatrix}, \qquad \det = -c_T V \neq 0 \iff V \neq 0.$$

$\Rightarrow$ T cells can be reconstructed as long as there is virus.

Identifiability (Theorem 2)

The lumped parameters $\Theta = [\alpha, \sigma, \gamma, r, \delta_T]$ can be recovered from $V, \dot{V}, \ddot{V}, \ldots$ ($\operatorname{rank} \partial\Phi/\partial\Theta = 5$).

Rule of thumb: if these fail, no observer will work, however well you tune it.

Step 3a · A clever change of coordinates (Eq. 13)

$$x_1 = \frac{V}{\lambda}, \qquad x_2 = \frac{\dot{V}}{\lambda} \qquad \Longrightarrow \qquad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- Both new states are **measured** (assuming $\dot{V}$ is available).
- T disappears from the equations. Solving (1) for it:

$$\boxed{c_T\, T = \alpha - \sigma V - \frac{\dot{V}}{V}} \qquad \Rightarrow \qquad \hat{T} = \frac{\alpha - \sigma \hat{V} - \hat{\dot{V}}/\hat{V}}{\hat{c}_T}$$

- $\lambda = 10^8$ is only a **scaling** factor: V reaches 10^9, so we work with $x_1 \sim 1$–10.

The model in the new coordinates (Eq. 14)

Differentiate x_2, replace T and $\dot{T}$ using (1)–(2) (see the handwritten derivation):

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{x_2^2}{x_1} + (\alpha\delta_T - \gamma)\, x_1 - \lambda\delta_T\sigma\, x_1^2 - r\,\alpha\, x_1\, G + \lambda\, r\,\sigma\, x_1^2\, G$$
$$- \delta_T\, x_2 - \lambda\sigma\, x_1 x_2 + r\, x_2\, G, \qquad G = G(\lambda x_1)$$

- Every term is a product of **known things**, **measured things** ($x_1 = y_1$, $x_2 = y_2$) and unknown parameters.
- Nothing in it depends on T. That is why we changed coordinates.

Step 3b · The state-affine form (Eq. 15): the recipe

$$\dot{x} = A(y)\,x + \beta(y) + \varphi(y)\,\theta, \qquad y = Cx$$

Sort every term of $\dot{x}$ into one of three bins:

1. **Contains an unknown parameter θ_i** $\rightarrow$ column i of $\varphi(y)$.
   The parameter must appear *linearly*: $\theta_i \times$ (function of y).
2. **Known coefficient $\times\, x_j$** $\rightarrow$ entry $(\cdot, j)$ of $A(y)$.
   Extra x's become data: $\lambda\sigma x_1^2 = (\lambda\sigma\, y_1)\, x_1$.
3. **Only data and known constants** $\rightarrow$ $\beta(y)$.

Golden rule

A, β, φ may only use **measured signals** y and **known constants**, never T and never θ.

Example 1 · Unknown: $\theta = \gamma = c_T S_T$

Regroup $\dot{x}_2$ (everything except γ is known):

$$\dot{x}_2 = \underbrace{\left[\alpha(\delta_T - rG) + \lambda\sigma y_1 (rG - \delta_T)\right]}_{a_{21}(y)} x_1 + \underbrace{\left[rG - \lambda\sigma y_1 - \delta_T\right]}_{a_{22}(y)} x_2 + \underbrace{\frac{y_2^2}{y_1}}_{\beta_2(y)} + \underbrace{(-y_1)}_{\varphi_2(y)} \gamma$$

$$A(y) = \begin{bmatrix} 0 & 1 \\ a_{21}(y) & a_{22}(y) \end{bmatrix}, \quad \beta(y) = \begin{bmatrix} 0 \\ y_2^2/y_1 \end{bmatrix}, \quad \varphi(y) = \begin{bmatrix} 0 \\ -y_1 \end{bmatrix}, \quad C = I_2$$

- Why γ and not c_T? $c_T \approx 2 \times 10^{-6}$ but $\gamma \approx 0.19$, and numbers of order 1 keep the solver happy. Then $\hat{c}_T = \hat{\gamma}/S_T$.
- Freedom: x_2^2/x_1 could also go to A as $(y_2/y_1)\, x_2$ (the paper's choice).

Example 2 · Unknowns: $\theta = [\gamma,\ r]^\top$

Now r is unknown too. Collect all the r-terms of (14) and factor out r:

$$-r\alpha x_1 G + \lambda r\sigma x_1^2 G + r x_2 G \;=\; r\, G(y)\big(y_2 - \alpha y_1 + \lambda\sigma y_1^2\big)$$

$$A(y) = \begin{bmatrix} 0 & 1 \\ \alpha\delta_T - \lambda\delta_T\sigma y_1 & -\delta_T - \lambda\sigma y_1 \end{bmatrix}, \qquad \beta(y) = \begin{bmatrix} 0 \\ y_2^2/y_1 \end{bmatrix}$$

$$\varphi(y) = \begin{bmatrix} 0 & 0 \\ -y_1 & G(\lambda y_1)\,\big(y_2 - \alpha y_1 + \lambda\sigma y_1^2\big) \end{bmatrix}, \qquad \theta = \begin{bmatrix} \gamma \\ r \end{bmatrix}$$

- The r-terms **moved from** A **to** φ. The observer equations do not change at all.
- Physical meaning: $y_2 - \alpha y_1 + \lambda\sigma y_1^2 = -c_T V T/\lambda$ (killing by T cells).

Step 4 · The adaptive observer (Eq. 17)

$$\dot{\hat{x}} = A\hat{x} + \beta + \varphi\hat{\theta} + \left\{S^{-1}C^\top + \Lambda\Gamma^{-1}\Lambda^\top C^\top\right\} Q\,(y - C\hat{x}) \qquad \text{copy of the model + correction}$$

$$\dot{S} = -\rho_x S - A^\top S - SA + C^\top QC \qquad \text{state gain (Riccati-like)}$$

$$\dot{\Lambda} = (A - S^{-1}C^\top QC)\Lambda + \varphi \qquad \text{sensitivity } \partial\hat{x}/\partial\hat{\theta}$$

$$\dot{\Gamma} = -\rho_\theta\Gamma + \Lambda^\top C^\top QC\Lambda \qquad \text{accumulated information}$$

$$\dot{\hat{\theta}} = \Gamma^{-1}\Lambda^\top C^\top Q\,(y - C\hat{x}) \qquad \text{parameter update}$$

- Sizes: $\hat{x} \in \mathbb{R}^n$, $S \in \mathbb{R}^{n\times n}$, $\Lambda \in \mathbb{R}^{n\times q}$, $\Gamma \in \mathbb{R}^{q\times q}$, $\hat{\theta} \in \mathbb{R}^q$. Here $n = 2$; $q = 1$ (Ex. 1) or $q = 2$ (Ex. 2).
- **Same five lines for any model** in state-affine form: only A, β, φ, C change.

How to read it

- **Output error** $e = y - C\hat{x}$: the only signal that "teaches" the observer.
- $\mathbf{\Lambda}$ answers: *"if $\hat{\theta}$ were a bit wrong, how would $\hat{x}$ move?"* It is driven by φ.
- $\mathbf{\Gamma}$ integrates $\Lambda^\top \Lambda$ with forgetting:

$$\Gamma(t) = e^{-\rho_\theta t}\Gamma(0) + \int_0^t e^{-\rho_\theta (t-s)}\, \Lambda^\top C^\top Q C \Lambda \, ds$$

  It measures **how much information we have collected** about θ.
- $\dot{\hat{\theta}} = \Gamma^{-1}\Lambda^\top C^\top Q\, e$: a gradient step, scaled by the inverse of that information (like recursive least squares).

Assumptions for convergence

A1 · Uniform observability of $\dot{x} = A(y)x,\ y = Cx$, with transition matrix Ψ:

$$\alpha_1 I \le \int_t^{t+T_1} \Psi^\top C^\top Q C\, \Psi\, ds \le \alpha_2 I \qquad \forall t \ge t_0$$

A2 · Stable sensitivity: with $L(t) = S^{-1}C^\top Q$ bounded,

$$\dot{\Lambda} = \big(A(y) - L(t)C\big)\Lambda \quad \text{is exponentially stable}$$

A3 · Persistent excitation (PE) of Λ, driven by φ (as in Eq. (4) of Aranovskiy et al. 2017):

$$\int_t^{t+T} \Lambda^\top C^\top Q C\, \Lambda\, ds \ \ge\ \mu\, I_q \quad \Longleftrightarrow \quad \lambda_{\min}\Big\{ \int_t^{t+T} \Lambda^\top C^\top Q C\, \Lambda\, ds \Big\} \ge \mu > 0 \qquad \forall t \ge t_0$$

Also: A, β, φ continuous and **bounded** along the trajectory.

If A1–A3 hold and ρ_x, ρ_θ are large enough, then $\hat{x} \to x$ and $\hat{\theta} \to \theta$ exponentially (Zhang 2002; Rodríguez et al. 2009, 2015).

The assumptions in plain words, for our examples

	Meaning	Our case
A1 observability	from y we can reconstruct x	$C = I$: we measure all of x ✓
A2 stable Λ	the correction $S^{-1}C^{\top}Q$ stabilises $A(y)$	choose ρ_x large (100) ✓
A3 PE	the data must *keep moving* so that θ shows its effect	$\varphi_{21} = -y_1 \neq 0$ only while the virus is high (days $\approx$2–10); see $\lambda_{\min}(\Gamma)$
bounded A, β, φ	no division by zero	needs $V > 0$: y_2/y_1, y_2^2/y_1 blow up if $V \to 0$

Lesson: **the observer can only learn while the infection is "exciting" the system.**

PE is a condition on the smallest eigenvalue

Compare with the gradient estimator for a linear regression $y = m^\top(t)\theta$ (Aranovskiy et al. 2017, Eq. 4):

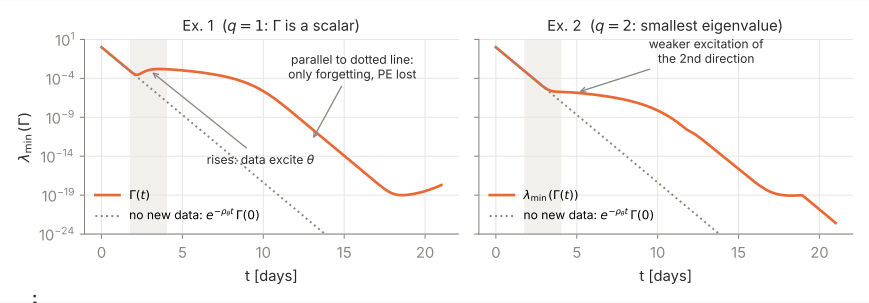
$$\underbrace{\int_t^{t+T} m(s)\, m^\top(s)\, ds \;\geq\; \mu I_q}_{\text{regression}} \qquad \longleftrightarrow \qquad \underbrace{\int_t^{t+T} \Lambda^\top C^\top Q C \Lambda\, ds \;\geq\; \mu I_q}_{\text{our observer (A3)}}$$

- Λ plays the role of the regressor m: it is φ **filtered** by the observer dynamics.
- "$\geq \mu I$" means **every eigenvalue** of the integral is $\geq \mu$, i.e. its minimal eigenvalue is $\geq \mu$.
- Meaning: in every window of length T, the data push $\hat{\theta}$ in **every direction** of $\mathbb{R}^q$.
  - Ex. 1 ($q = 1$): the integral is a scalar, $\int \Lambda^\top \Lambda\, ds > 0$: Λ must not stay at 0.
  - Ex. 2 ($q = 2$): exciting γ *or* r is not enough; the **least** excited direction decides.

An online PE meter: $\lambda_{\min}(\Gamma(t))$

$\Gamma(t)$ is the PE integral **with exponential forgetting**, and it is already inside the observer:

$$\Gamma(t) = e^{-\rho_\theta t}\Gamma(0) + \int_0^t e^{-\rho_\theta (t-s)}\,\Lambda^\top C^\top Q C \Lambda\, ds \quad \Rightarrow \quad \text{PE} \implies \lambda_{\min}(\Gamma(t)) \geq e^{-\rho_\theta T}\mu$$



- Since $\dot{\hat{\theta}} = \Gamma^{-1}\Lambda^\top C^\top Q\, e$ and $\|\Gamma^{-1}\| = 1/\lambda_{\min}(\Gamma)$: when PE is lost, the gain explodes and noise drives $\hat{\theta}$.

Tuning knobs

Knob	Value used	Effect when you increase it
ρ_x	100	faster state correction, more sensitive to noise
ρ_θ	4	faster forgetting $\Rightarrow$ faster adaptation, noisier $\hat{\theta}$
Q	I_2	more weight on the output error
$\Gamma(0)$	I_q	larger $\Rightarrow$ smaller initial gain (slow start)
$S(0)$, $\Lambda(0)$	I_2, 0	usually not critical
$\hat{x}(0)$, $\hat{\theta}(0)$	$0.5\,y(0)$; 0.05	initial guesses (true: $\gamma = 0.189$, $r = 0.194$)
λ	10^8	scaling only: keeps x_1 of order 1–10

From equations to code

One ODE for **patient + observer**.

State vector (Ex. 1, 11 entries):

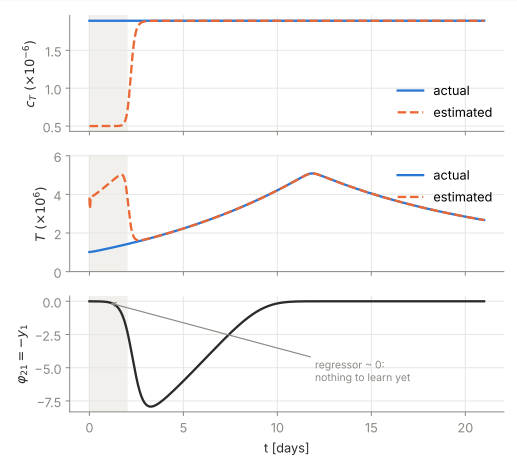
- `z(1:2)` V, T (patient)
- `z(3:4)` $\hat{x}_1$, $\hat{x}_2$
- `z(5:7)` S_{11}, S_{12}, S_{22}
- `z(8:9)` Λ
- `z(10)` Γ
- `z(11)` $\hat{\gamma}$

Files: `Ex1_main_cT.m`, `Ex1_AdapObs_cT.m`, `patient_model.m` (and `ex1_cT.py`)

```
% measured data -> state-affine matrices
y1 = V/lambda;  y2 = dV/lambda;
A    = [0 1; a21 a22];
beta = [0; y2^2/y1];
phi  = [0; -y1];
C    = eye(2);

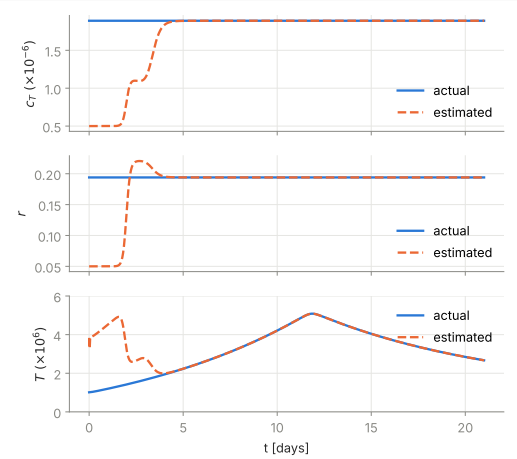
% adaptive observer, Eq. (17)
y_err  = y - C*x_hat;
dx_hat = A*x_hat + beta + phi*theta_hat ...
   + (S\C' + Lambda*(Gamma\Lambda')*C')*Q*y_err;
dS      = -rho_x*S - A'*S - S*A + C'*Q*C;
dLambda = (A - S\C'*Q*C)*Lambda + phi;
dGamma  = -rho_theta*Gamma + Lambda'*C'*Q*C*Lambda;
dtheta_hat = Gamma\Lambda'*C'*Q*y_err;
```

Results · Example 1 (estimate c_T)



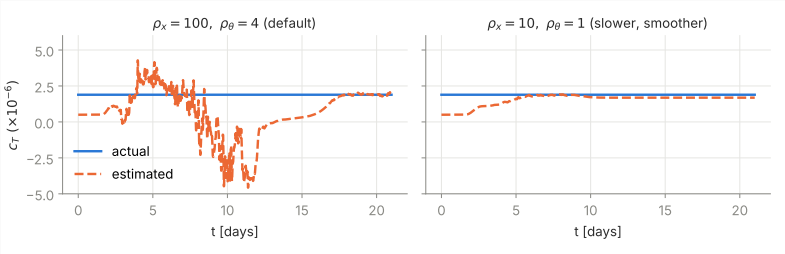
- $\hat{c}_T$ reaches the true value by **day** $\approx$ **4**.
- $\hat{T}$ is wrong while $\hat{c}_T$ is wrong, then follows T.
- **Shaded:** the first two days, when $\varphi_{21} = -y_1 \approx 0$. There is no excitation, so $\hat{c}_T$ cannot move (A3).

Results · Example 2 (estimate c_T and r)



- Both parameters converge by **day $\approx$ 6**.
- One more unknown means you need more excitation, so convergence is slower (as in Table I of the paper).
- Same code: only A, φ and the sizes of $\Lambda, \Gamma, \hat{\theta}$ changed.

With real data: noise, sampling, detection limits



Example 2, 5% multiplicative noise on V. Left: default tuning. Right: smaller ρ_x, ρ_θ (more memory).

- **Noise:** the drift (days 7–12) appears when $\lambda_{\min}(\Gamma)$ is tiny. Trade speed for robustness with ρ_x, ρ_θ.
- $\dot{V}$ **is not measured** in the clinic; it has to be estimated with numerical differentiators.
- **Detection limit:** below 50 copies/ml, $V \approx 0$, so the $1/y_1$ terms and PE are both lost: stop adapting.

Hands-on exercises

1. **Run** Ex. 1 and Ex. 2. Change the initial guesses $\hat{\theta}(0)$ and $\Gamma(0)$. What changes?
2. **Other patients:** use the parameter sets A–I of Hernandez-Vargas (2020). Which ones converge slowest, and why? (Hint: look at φ.)
3. **Noise:** set `add_noise = true`. Re-tune ρ_x and ρ_θ.
4. **Start late:** start the observer at day 12. Is there still PE? Look at the PE meter $\lambda_{\min}(\Gamma(t))$.
5. **Your turn:** estimate $\theta = [\gamma,\ \delta_T]$ instead. Find A, β, φ.
   Hint: the δ_T-column of φ is $\alpha y_1 - \lambda\sigma y_1^2 - y_2$.

Take-home recipe

1. **Model**: what is measured (y), what is unknown (x, θ)?
2. **Check** observability and identifiability.
3. **Change coordinates** so the unknown states disappear and the parameters enter *linearly*.
4. **Sort terms** into $A(y)$, $\beta(y)$, $\varphi(y)\theta$, using only measured data.
5. **Copy Eq. (17)**, check A1–A3 (above all, persistent excitation), tune ρ_x, ρ_θ.

Thank you!

Code (MATLAB & Python) in the course folder.

References

- A. Rodríguez-Liñán, J. E. Sereno, G. Quiroz-Compean, E. A. Hernandez-Vargas. *Adaptive observers for calibrating immune digital twins in viral infections.* IEEE CDC, 2025.
- E. A. Hernandez-Vargas, J. X. Velasco-Hernandez. *In-host mathematical modelling of COVID-19 in humans.* Annual Reviews in Control 50:448–456, 2020.
- Q. Zhang. *Adaptive observer for MIMO linear time-varying systems.* IEEE TAC 47(3):525–529, 2002.
- A. Rodríguez, J. De León, R. Femat, C. Hernández-Rosales. *A dynamic parameter estimator to control chaos with distinct feedback schemes.* CNSNS 14(12):4280–4291, 2009.
- A. Rodríguez, G. Quiroz, R. Femat, H. Méndez-Acosta, J. de León. *An adaptive observer for operation monitoring of anaerobic digestion wastewater treatment.* Chem. Eng. J. 269:186–193, 2015.
- J. de León Morales, A. Chaillet, A. Loría, G. Besançon. *Output feedback control via adaptive observers with persistency of excitation.* IFAC 2005.
- S. Aranovskiy et al. *Performance enhancement of parameter estimators via dynamic regressor extension and mixing.* IEEE TAC, 2017.
- R. Visentin, C. Dalla Man, Y. C. Kudva, A. Basu, C. Cobelli. *Circadian variability of insulin sensitivity: physiological input for in silico artificial pancreas.* Diabetes Technol. Ther. 17(1):1–7, 2015.
- C. Ramírez Mazo, L. Gómez-Echavarría, L. Lema-Pérez, S. Röblitz. *A physiologically based mathematical model describing estradiol and progesterone actions on glucose-insulin dynamics throughout the menstrual cycle.* J. Theor. Biol. 631:112518, 2026.

Open projects: PAE · undergrad thesis · master thesis

1 · Circadian insulin sensitivity in T1D

Objective. Design an adaptive observer that tracks the time-varying insulin sensitivity $S_I(t)$ of a person with T1D from CGM and insulin-pump data, as in the breakfast/lunch/dinner pattern of Fig. 1 in Visentin et al. (2015).

Methodology.

1. Glucose–insulin minimal model with S_I as unknown parameter.
2. Rewrite it in state-affine form; check observability, identifiability and PE (meals as excitation).
3. Generate in silico data with a circadian S_I profile.
4. Tune ρ_θ to follow S_I changes; test noise and 5-min CGM sampling.

2 · Menstrual cycle and diabetes

Objective. Estimate the cycle-dependent insulin sensitivity $S_I(E_2, P_4)$ along the menstrual cycle of women with diabetes, based on the model of Ramírez Mazo et al. (2026).

Methodology.

1. Adapt the G–I part of the model to diabetes (exogenous insulin input).
2. Choose the unknown parameters of $S_I(E_2, P_4)$ and write the state-affine form.
3. Analyse observability, identifiability and PE over a 28-day cycle.
4. Validate the observer in silico with E_2, P_4 profiles and CGM data.